

Customer Churn Prediction Using Machine Learning

Project Report

1. Introduction

Customer churn is an important problem for telecommunication companies because losing existing customers can directly affect revenue. The aim of this project is to build a machine learning model that can predict whether a customer is likely to leave the service. The project follows a complete workflow starting from data exploration and cleaning, followed by feature engineering and machine learning model development.

2. Dataset and Exploratory Data Analysis

The project uses a telecommunications customer dataset containing 7,043 customer records and 21 original columns. The dataset includes information about customer demographics, tenure, phone and internet services, contracts, payment methods and billing. The target variable is Churn. During the exploratory analysis, the numerical variables were examined using summary statistics. The average customer tenure was approximately 32.37 months, while the average monthly charge was about USD 64.76. The analysis also included checking data types, missing values and the distribution of different customer attributes.

3. Data Cleaning

One of the main cleaning steps was converting the TotalCharges column from a text field into a numeric field. After conversion, 11 missing values were found in TotalCharges. Since this represented only a very small part of the dataset, those records were removed before further analysis.

4. Feature Engineering

Additional features were created to provide the models with more meaningful information about customer behaviour. CustomerID was removed because it does not provide useful predictive information. Churn was converted from Yes/No to a binary value of 1/0.

- TenureGroup: groups customers according to their length of service.
- AvgMonthlySpend: estimates the customer's average monthly spending using TotalCharges and tenure.
- NumAddOnServices: counts the additional services used by a customer.
- HasInternet: indicates whether the customer has an internet service.

5. Model Development

The data was divided into training and testing sets. Numerical variables were standardized and categorical variables were converted into numerical form using one-hot encoding. Several classification algorithms were then compared using stratified five-fold cross-validation and ROC-AUC.

Model	Mean CV ROC-AUC
Logistic Regression	0.8477

Model	Mean CV ROC-AUC
Random Forest	0.8438
Gradient Boosting	0.8481
XGBoost	0.8447
LightGBM	0.8368

Gradient Boosting gave the highest mean ROC-AUC among the baseline models. Grid-search tuning was then performed for the Gradient Boosting model. The selected configuration used a learning rate of 0.1, maximum tree depth of 2 and 100 estimators, giving a cross-validated ROC-AUC of approximately 0.8518.

6. Handling Class Imbalance

The project also investigated class imbalance using SMOTEENN. This technique combines oversampling of the minority class with cleaning of difficult or noisy samples. A Gradient Boosting classifier was trained using the resampled training data.

7. Final Results

The reported SMOTEENN-based Gradient Boosting evaluation produced strong classification performance. On the 1,317-row evaluation set, the model achieved an accuracy of 94.46%. For the churn class, precision was 0.94, recall was 0.96 and the F1-score was 0.95.

Metric	Result
Accuracy	94.46%
Precision (Churn)	0.94
Recall (Churn)	0.96
F1-score (Churn)	0.95
Macro F1-score	0.94

The confusion matrix for this evaluation was [[520, 45], [28, 724]]. This means that 520 non-churn customers and 724 churn customers were correctly classified, while 45 non-churn customers were incorrectly classified as churn and 28 churn customers were missed.

8. Conclusion

This project demonstrates a complete machine learning approach for customer churn prediction. The work began with understanding and cleaning the customer data, followed by creating useful features and preparing the data for machine learning. Several classification models were compared, with Gradient Boosting giving the strongest baseline ROC-AUC. Hyperparameter tuning further improved its cross-validation performance.

The SMOTEENN experiment produced a reported accuracy of 94.46% and an F1-score of 0.95 for the churn class. Overall, the project shows that machine learning can be used effectively to identify customers who are more likely to churn and can therefore support customer-retention decisions.

9. Key Points

- The project uses 7,043 customer records.

- TotalCharges was converted to numeric and 11 incomplete records were removed.
- Four useful features were created during feature engineering.
- Five machine learning classifiers were compared.
- Gradient Boosting achieved the best baseline cross-validation ROC-AUC.
- SMOTEENN was used to address class imbalance.
- The final reported SMOTEENN evaluation achieved 94.46% accuracy and 0.95 churn F1-score.